Summary of Cyber-Physical Systems (CPS) Research and Programs

1. General Frameworks and Programs

- **NIST CPS & IoT Program:** The National Institute of Standards and Technology (NIST) program focuses on converging the definitions of CPS and Internet of Things (IoT) to support hybrid systems integrating digital, physical, and human components. The program's objective is to enable scalable, dependable design methods and reproducible performance metrics for trustworthy systems (reliable, safe, secure, resilient). It highlights the economic potential of next-generation CPS in sectors like transportation and energy, while addressing critical challenges in scalability, interoperability, and the lack of existing measurement science.

2. CPS Architectures and Platforms

- **The AXIOM Platform:** Two documents detail AXIOM (Agile, eXtensible, fast I/O Module), a hardware-software platform designed for next-generation CPS.
  - **Goal:** To provide a scalable, easy-to-program platform that bridges the gap between heterogeneous hardware and software design.
  - **Architecture:** It combines general-purpose processors (like ARM) with programmable logic (FPGAs) to accelerate specific tasks. It features a high-speed, low-cost interconnect called "AXIOM-Link" to scale performance by connecting multiple boards.
  - **Programming:** AXIOM uses the OmpSs task-based parallel programming model, allowing developers to annotate code for execution on either standard processors or FPGA accelerators.
  - **Use Cases:** The platform is validated through applications such as smart video surveillance and smart home living.

3. Dependability and Monitoring

- **CPS-IoT Dependability:** One paper addresses the reliability of CPS infrastructures that rely on IoT subsystems.
  - **Methodology:** It proposes an orchestration model featuring "inbound surveillance" to monitor network integrity and "outbound actuation" to control operations.
  - **Epidemic Modeling:** The authors adapt mathematical epidemiology (SIR/SAR models) to model information diffusion and congestion (interference) within IoT networks, translating "infected" states into network congestion or failure nodes.
  - **Protocol:** A novel protocol called "INMP" (Interference-aware clustering with Node Migration) is introduced to reconfigure networks dynamically, preventing routing oscillations and improving safety.

4. Applications: Smart Cities

- **Smart Cities in India:** This document analyzes the role of CPS in transforming Indian cities into smart cities.
    - **Key Action Areas:** It identifies core components such as Smart Economy, Smart Governance, Smart Environment, Smart Mobility, and Smart Living.
    - **Enabling Technologies:** The paper highlights the necessity of IoT, Artificial Intelligence (AI), Blockchain, and 5G/SDN as the backbone for these initiatives.
    - **Challenges:** It discusses specific Indian challenges, such as managing unplanned legacy cities, diverse populations, and the high cost of infrastructure. Security threats to actuators and data management systems are also a major concern.

5. Security

- **CPS Security Survey:** A comprehensive survey reviews security issues across multiple CPS domains, including Industrial Control Systems (ICS), Smart Grids, Medical Devices, and Smart Cars.
    - **Framework:** The authors propose a unified framework analyzing security across three coordinates: security perspective (threats, vulnerabilities, attacks, controls), CPS components (cyber, physical, cyber-physical), and system types.
    - **Specific Threats:** The survey details domain-specific risks, such as "false data injection" in smart grids, jamming attacks on implantable medical devices, and remote compromise of automotive Electronic Control Units (ECUs).
    - **Challenges:** It identifies key challenges like the lack of "security by design," the difficulty of securing legacy systems, and the privacy risks associated with granular data collection (e.g., smart meters).